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Roll No 02

Implementation of a Battery-Aware Simple Reflex Vacuum Cleaner Agent

Aim

To implement a Simple Reflex Vacuum Cleaner Agent in Python that cleans different areas of a college library based on their current cleanliness condition and evaluates its performance using a scoring mechanism.

Theory

Simple Reflex Agent

A **Simple Reflex Agent** is an AI agent that selects an action based only on the **current percept** received from the environment. It does not use past experiences or maintain a history of previous states.

In this implementation, a vacuum cleaner operates in different areas of a college library:

- **Reading Hall**
- **Book Section**
- **Reception**
- **Cabin 1**
- **Cabin 2**

Each area can have one of two states:

- **0 → Clean**
- **1 → Dirty**

Condition-Action Rules

Current Percept	Action	Performance
Area is Dirty	Clean the area	+1
Area is Clean	No Action	0
Move to next area	Move	-1
Battery is Low	Go to charging station	Battery restored

The initial condition of each library area is generated randomly using Python's `random` module.

The vacuum cleaner also has a **battery level**. Cleaning and movement consume battery power. When the battery becomes low, the agent goes to the charging station and restores its battery to 100%.

The **performance score** measures how efficiently the vacuum cleaner performs its task. Cleaning provides a positive reward, while movement has a negative cost.

Components of the Agent

Environment:
Represents the college library and stores the cleanliness condition of each area.

Agent:
Represents the vacuum cleaner and decides which action to perform.

Percept:
The current cleanliness condition of the area and the battery condition.

Actions:
The vacuum cleaner can clean, move, or charge.

Performance Measure:
The agent receives `+1` for cleaning a dirty area and `-1` for moving to another area.

Flowchart

```
              ┌──────────────┐
              │    START     │
              └──────┬───────┘
                     │
                     ▼
         ┌──────────────────────┐
         │ Create Library       │
         │ Environment          │
         └──────────┬───────────┘
                    │
                    ▼
         ┌──────────────────────┐
         │ Randomly assign      │
         │ Clean / Dirty states │
         └──────────┬───────────┘
                    │
                    ▼
         ┌──────────────────────┐
         │ Create Vacuum Agent  │
```

```
│ Battery = 100%           │
│ Score = 0                │
└────────────┬─────────────┘
             │
             │
             ▼
   ┌──────────────────┐
   │ Select Current   │
   │ Library Area     │
   └────────┬─────────┘
            │
            │
            ▼
     ┌──────────────┐
     │ Battery Low? │
     └──────┬───────┘
        Yes │ No
            │
            ▼
    ┌──────────────┐
    │    Charge    │
    │ Battery 100% │
    └──────┬───────┘
           │
           │
           └──────────────┐
                          │
                          ▼
                  ┌──────────────┐
                  │ Area Dirty?  │
                  └──────┬───────┘
                     Yes │ No
                         │
           ┌─────────────┴─────────────┐
           │                           │
           ▼                           ▼
   ┌──────────────┐            ┌──────────────┐
   │    Clean     │            │  No Action   │
   │    Score +1  │            │   Score 0    │
   └──────┬───────┘            └──────┬───────┘
          │                           │
          └─────────────┬─────────────┘
                        │
                        ▼
                ┌─────────────────┐
                │ More Areas?     │
                └───────┬─────────┘
                    Yes │ No
                        │
```

```
                ┌───────────┴───────────┐
                │                       │
                ▼                       ▼
        ┌──────────────┐         ┌───────────┐
        │ Move to Next │         │   STOP    │
        │ Area Score-1 │         └───────────┘
        └──────┬───────┘
               │
               └───────────────►
                      Next Area
```

```python
import random




# ==========================================================
# ENVIRONMENT CLASS
# Represents the MITAOE college library
# ==========================================================

class Environment:

    def __init__(self):

        # 0 = Clean
        # 1 = Dirty

        # The initial condition of every area is random.
        # This makes every run slightly different.
        self.rooms = {
            "Reading Hall": random.randint(0, 1),
            "Book Section": random.randint(0, 1),
            "Reception": random.randint(0, 1),
            "Cabin 1": random.randint(0, 1),
            "Cabin 2": random.randint(0, 1)
        }

    # ------------------------------------------------------
    # Returns the current condition of a particular area
    # ------------------------------------------------------
    def get_status(self, room):
        return self.rooms[room]

    # ------------------------------------------------------
    # Cleans a dirty area
    # ------------------------------------------------------
    def clean_room(self, room):
        self.rooms[room] = 0

    # ------------------------------------------------------
    # Displays the current condition of the library
```

```python
    # -----------------------------------------------------
    def display(self):

        print("\nCurrent Library Environment:")

        for room, status in self.rooms.items():

            if status == 1:
                print(room, "-> Dirty")
            else:
                print(room, "-> Clean")


# ==========================================================
# SIMPLE REFLEX AGENT CLASS
# Represents the automatic vacuum cleaner
# ==========================================================

class ReflexAgent:

    def __init__(self, environment):

        # Connect the vacuum cleaner with the environment
        self.environment = environment

        # The vacuum starts from the Reading Hall
        self.location = "Reading Hall"

        # Initial performance score
        self.score = 0

        # Battery starts fully charged
        self.battery = 100

        # Number of areas cleaned by the vacuum
        self.cleaned_rooms = 0

        # Charging station is assumed to be near Reception
        self.charging_station = "Reception"

    # -----------------------------------------------------
    # Charges the vacuum when its battery is low
    # -----------------------------------------------------
    def charge(self):

        print("\nBattery is low!")
        print("Vacuum is going to the charging station...")

        # Move the vacuum to the charging station
        self.location = self.charging_station

        # Restore battery
        self.battery = 100

        print("Battery fully charged: 100%")
```

```
    # -------------------------------------------------------
    # Main decision-making method of the reflex agent
    # -------------------------------------------------------
    def act(self):

        # Order in which the vacuum visits the library
        locations = [
            "Reading Hall",
            "Book Section",
            "Reception",
            "Cabin 1",
            "Cabin 2"
        ]

        # Visit each area one by one
        for i in range(len(locations)):

            # Update the current location
            self.location = locations[i]

            print("\n----------------------------------------")
            print("Vacuum is at:", self.location)
            print("Battery:", self.battery, "%")
            print("Score:", self.score)

            # -------------------------------------------------
            # Check battery before performing any action
            # -------------------------------------------------

            if self.battery <= 20:

                self.charge()

            # -------------------------------------------------
            # Check whether the current area is dirty
            # -------------------------------------------------

            status = self.environment.get_status(self.location)

            # If the area is dirty
            if status == 1:

                print("Percept: Area is Dirty")
                print("Action: Cleaning the area...")

                # Clean the area
                self.environment.clean_room(self.location)

                # Cleaning gives a reward
                self.score += 1

                # Cleaning consumes battery
                self.battery -= 2
```

```
                # Increase number of cleaned areas
                self.cleaned_rooms += 1

                print("Area cleaned successfully.")

            # If the area is already clean
            else:

                print("Percept: Area is Clean")
                print("Action: No cleaning required.")

            # -------------------------------------------------
            # Move to the next area
            # -------------------------------------------------

            if i < len(locations) - 1:

                print("Moving to the next area...")

                # Movement has a cost
                self.score -= 1

                # Movement consumes battery

                self.battery -= 1

        # FINAL RESULT

        print("\n========================================")
        print("       CLEANING COMPLETED")
        print("========================================")

        print("\nFinal Library Condition:")
        self.environment.display()

        print("\n---------- Agent Performance ----------")
        print("Areas cleaned:", self.cleaned_rooms)
        print("Remaining battery:", self.battery, "%")
        print("Final performance score:", self.score)



# MAIN PROGRAM

# Create the library environment
environment = Environment()

# Show the initial condition of the library
print("========================================")
print("      MITAOE LIBRARY ENVIRONMENT")
print("========================================")

environment.display()

# Create the vacuum cleaner agent
```

```
agent = ReflexAgent(environment)

# Start the vacuum cleaner
print("\n========================================")
print("        VACUUM AGENT STARTS")
print("========================================")

agent.act()
```

```
========================================
      MITAOE LIBRARY ENVIRONMENT
========================================

Current Library Environment:
Reading Hall -> Clean
Book Section -> Clean
Reception -> Clean
Cabin 1 -> Clean
Cabin 2 -> Dirty

========================================
        VACUUM AGENT STARTS
========================================

----------------------------------------
Vacuum is at: Reading Hall
Battery: 100 %
Score: 0
Percept: Area is Clean
Action: No cleaning required.
Moving to the next area...

----------------------------------------
Vacuum is at: Book Section
Battery: 99 %
Score: -1
Percept: Area is Clean
Action: No cleaning required.
Moving to the next area...

----------------------------------------
Vacuum is at: Reception
Battery: 98 %
Score: -2
Percept: Area is Clean
Action: No cleaning required.
Moving to the next area...

----------------------------------------
Vacuum is at: Cabin 1
Battery: 97 %
Score: -3
Percept: Area is Clean
Action: No cleaning required.
Moving to the next area...

----------------------------------------
Vacuum is at: Cabin 2
Battery: 96 %
```

```
Score: -4
Percept: Area is Dirty
Action: Cleaning the area...
Area cleaned successfully.

========================================
       CLEANING COMPLETED
```